Experiment No.09

Aim: Implement Recurrent Neural Network (RNN) / LSTM for time series or text data.

Theory:

1. Dataset Source

Dataset: Air Quality Dataset

Source Link:
https://archive.ics.uci.edu/dataset/360/air+quality

2. Dataset Description

The Air Quality dataset contains hourly averaged responses from a gas multisensor device deployed in an Italian city to monitor environmental pollution.

Key Details:

- Total records: 9358 observations
- Type: Multivariate time series dataset
- Frequency: Hourly measurements

Features:

- Pollutants: CO(GT), NOx(GT), NO2(GT), Benzene (C6H6)
- Sensor readings: PT08 series
- Environmental factors:
    - Temperature (T)
    - Relative Humidity (RH)
    - Absolute Humidity (AH)

Target Variable:

- Temperature (T) used for prediction (can also use CO levels)

Data Characteristics:

- Missing values represented as -200
- Continuous numerical data
- Strong temporal dependency

3. Mathematical Formulation of LSTM

LSTM is an advanced RNN architecture designed to capture long-term dependencies in sequential data.

Forget Gate

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$

Input Gate

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$$

Candidate State

$$\tilde{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c)$$

Cell State Update

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$$

Output Gate

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o)$$

Hidden State

$$h_t = o_t \cdot \tanh(C_t)$$

4. Algorithm Limitations

- Computationally expensive due to complex architecture
- Requires large datasets for better generalization
- Training is time-consuming
- Sensitive to hyperparameters (units, window size)
- Can overfit if not regularized

- Performance affected by noisy or missing data

5. Methodology / Workflow

Steps:

1. Load AirQuality dataset into MATLAB
2. Extract numeric columns
3. Replace invalid values (-200) with NaN
4. Apply interpolation to fill missing values
5. Select temperature (T) as target variable
6. Normalize data to [0,1] range
7. Create sequences using sliding window (24 time steps)
8. Split dataset into training and testing sets
9. Build LSTM model:
    - Input layer
    - LSTM layer
    - Dropout layer
    - Fully connected layer
10. Train model using Adam optimizer
11. Predict future values
12. Evaluate using RMSE and MAE
13. Visualize:

- Raw data graph
- Forecast graph

6. Performance Analysis

Evaluation Metrics:

- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)

Observations:

- Predicted values closely follow actual temperature trends
- Model captures seasonal and cyclic patterns
- Slight smoothing observed in predictions
- Minor deviations occur at sharp peaks

Interpretation:

- LSTM effectively learns temporal dependencies
- Good forecasting capability for environmental data
- Lower RMSE indicates better model accuracy

7. Hyperparameter Tuning

Parameters Tuned:

Parameter	**Values**
Window Size	12, 24
LSTM Units	32, 64, 100
Epochs	50, 60, 100
Batch Size	32, 64
Dropout	0.2

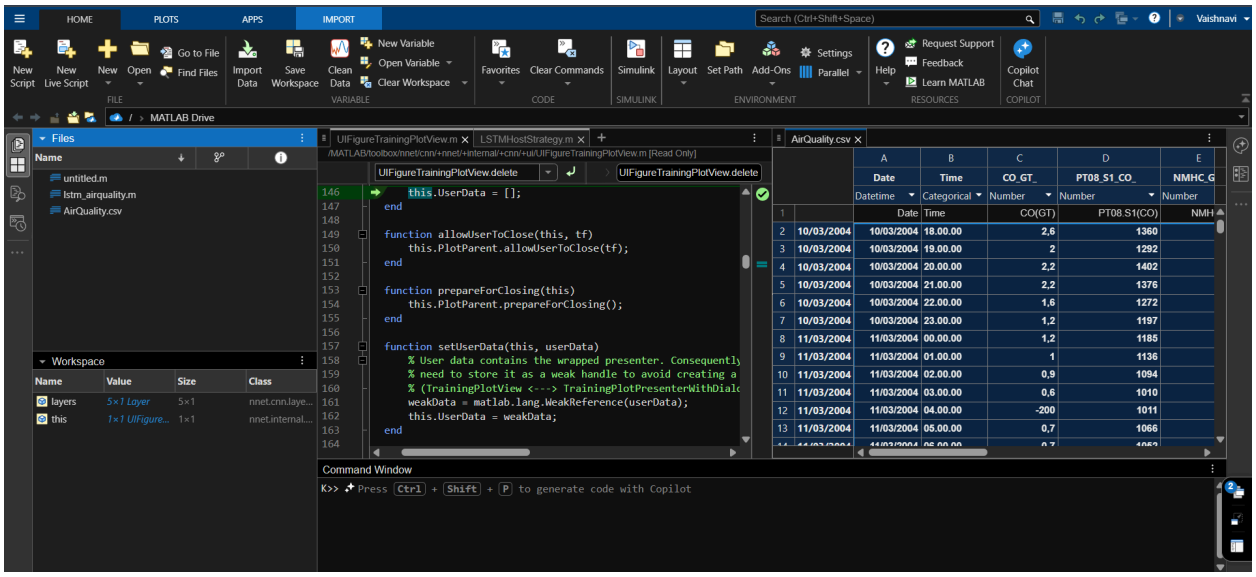
Final Configuration Used:

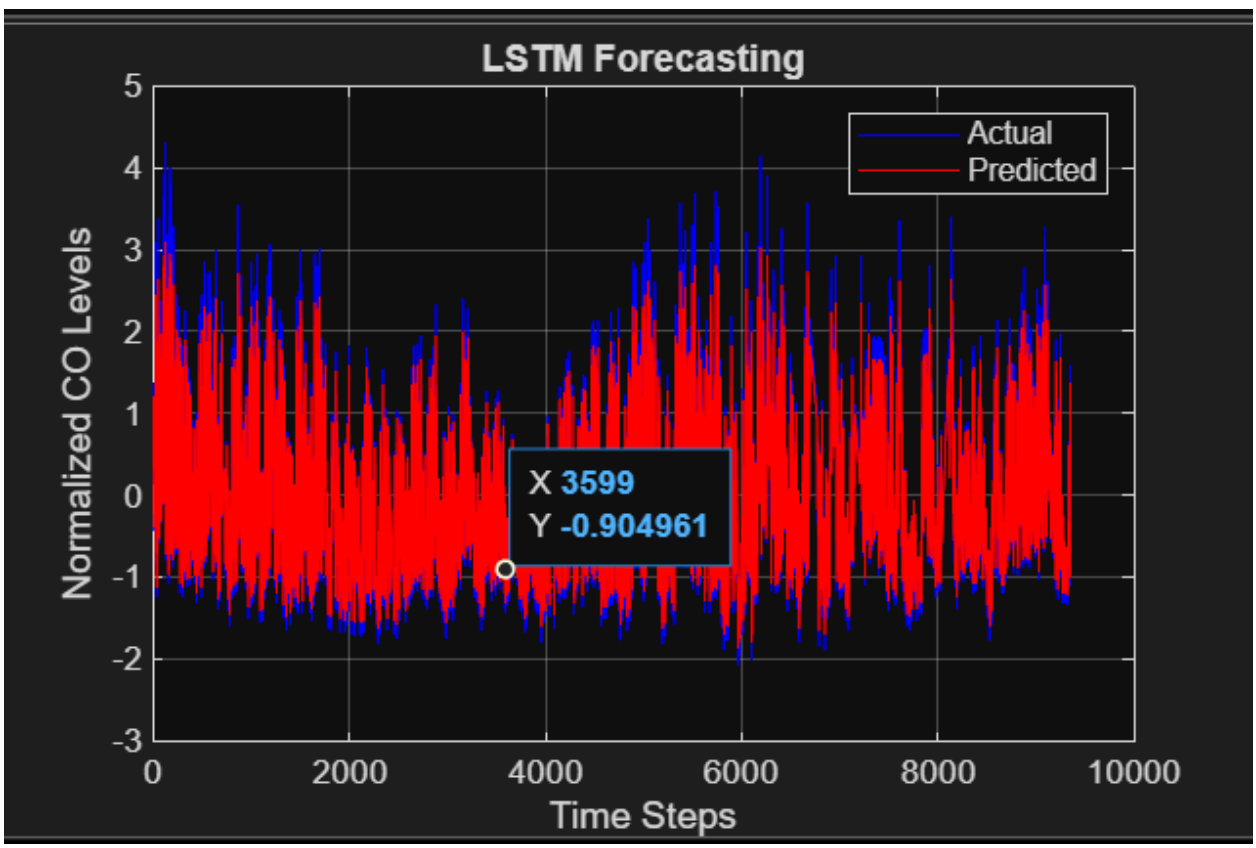
- Window size: 24
- LSTM units: 64
- Epochs: 60
- Batch size: 64
- Dropout: 0.2

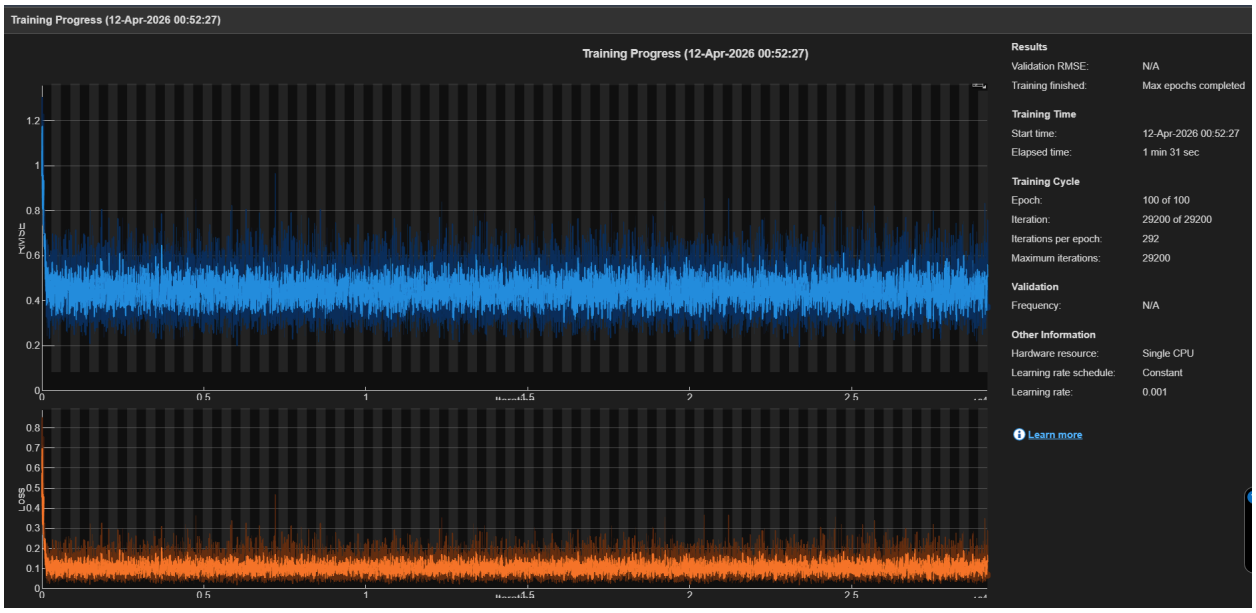
Impact of Tuning:

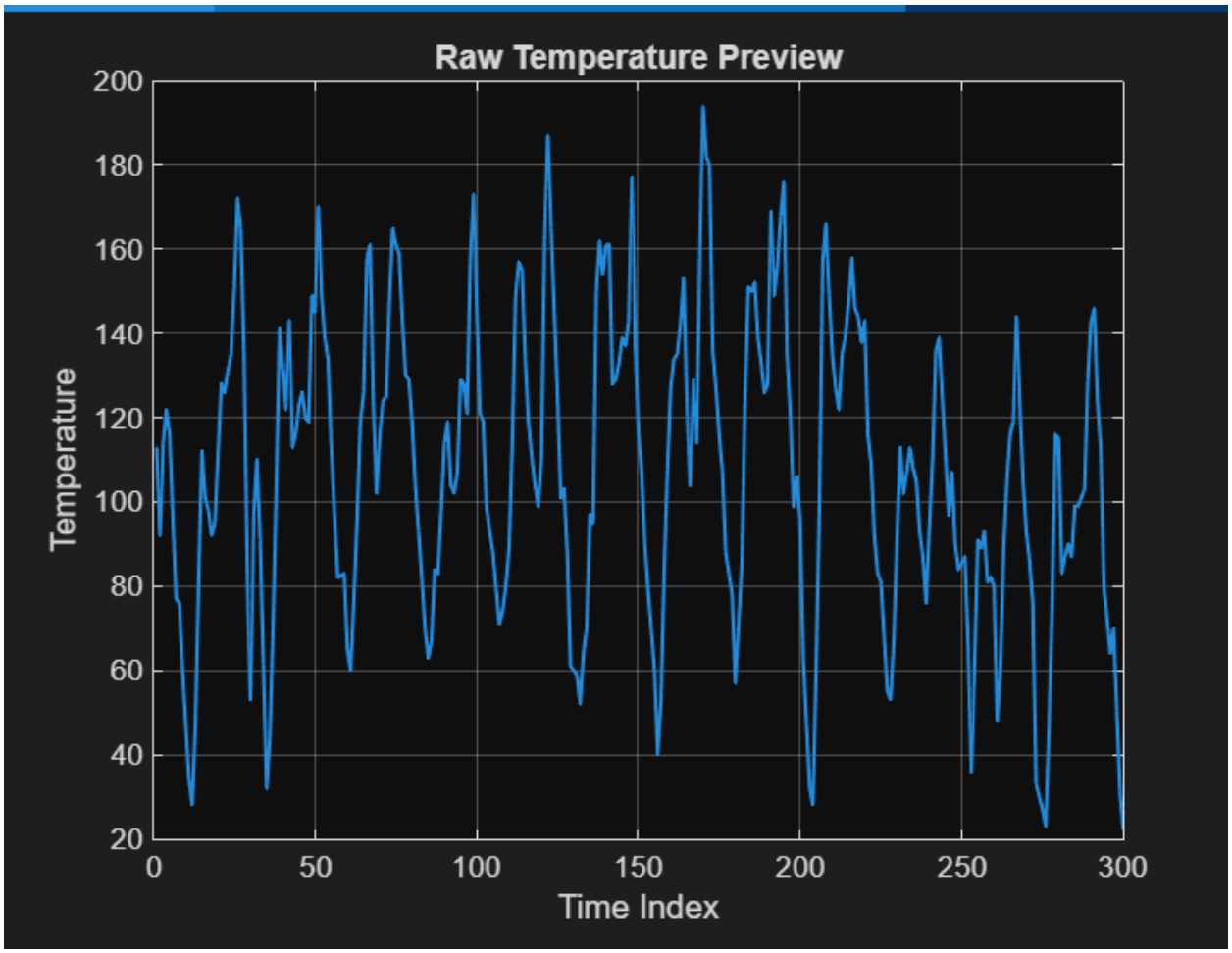
- Improved model stability
- Reduced overfitting
- Better prediction accuracy

Output:


HOME
PLOTS
APPS
IMPORT
Search (Ctrl+Shift+Space)
Vaishnavi
New Script
New Live Script
New
Open
Go to File
Find Files
Import Data
Save Workspace
Clean Data
New Variable
Open Variable
Clear Workspace
Favorites
Clear Commands
Simulink
Layout
Set Path
Add-Ons
Settings
Parallel
Help
Request Support
Feedback
Learn MATLAB
Copilot Chat
FILE
VARIABLE
CODE
SIMULINK
ENVIRONMENT
RESOURCES
COPILOT
MATLAB Drive
Files
Name
untitled.m
lstm_airquality.m
AirQuality.csv
Workspace
Name
Value
Size
Class
layers
5×1 Layer
5×1
nnet.cnn.laye...
this
1×1 UIFigure...
1×1
nnet.internal....
UIFigureTrainingPlotView.m
LSTMHostStrategy.m
/MATLAB/toolbox/nnet/cnn/+nnet/+internal/+cnn/+ui/UIFigureTrainingPlotView.m [Read Only]
UIFigureTrainingPlotView.delete
this.UserData = [];
end
function allowUserToClose(this, tf)
this.PlotParent.allowUserToClose(tf);
end
function prepareForClosing(this)
this.PlotParent.prepareForClosing();
end
function setUserData(this, userData)
% User data contains the wrapped presenter. Consequently
% need to store it as a weak handle to avoid creating a
% (TrainingPlotView <---> TrainingPlotPresenterWithDialo
weakData = matlab.lang.WeakReference(userData);
this.UserData = weakData;
end
AirQuality.csv
Date
Time
CO_GT_
PT08_S1_CO_
NMHC_G
Datetime
Categorical
Number
Date
Time
CO(GT)
PT08.S1(CO)
NMH
10/03/2004 18.00.00 2,6 1360
10/03/2004 19.00.00 2 1292
10/03/2004 20.00.00 2,2 1402
10/03/2004 21.00.00 2,2 1376
10/03/2004 22.00.00 1,6 1272
10/03/2004 23.00.00 1,2 1197
11/03/2004 00.00.00 1,2 1185
11/03/2004 01.00.00 1 1136
11/03/2004 02.00.00 0,9 1094
11/03/2004 03.00.00 0,6 1010
11/03/2004 04.00.00 -200 1011
11/03/2004 05.00.00 0,7 1066
Command Window
K>> Press Ctrl + Shift + P to generate code with Copilot


LSTM Forecasting
Actual
Predicted
X 3599
Y -0.904961
Normalized CO Levels
Time Steps
5
4
3
2
1
0
-1
-2
-3
0
2000
4000
6000
8000
10000


Training Progress (12-Apr-2026 00:52:27)
Training Progress (12-Apr-2026 00:52:27)
RMSE
Loss
Results
Validation RMSE: N/A
Training finished: Max epochs completed
Training Time
Start time: 12-Apr-2026 00:52:27
Elapsed time: 1 min 31 sec
Training Cycle
Epoch: 100 of 100
Iteration: 29200 of 29200
Iterations per epoch: 292
Maximum iterations: 29200
Validation
Frequency: N/A
Other Information
Hardware resource: Single CPU
Learning rate schedule: Constant
Learning rate: 0.001
Learn more


Raw Temperature Preview
Temperature
Time Index
200
180
160
140
120
100
80
60
40
20
0
50
100
150
200
250
300

Conclusion:

The LSTM model successfully captures temporal patterns in air quality data and provides accurate predictions for future values. The model demonstrates strong capability in handling time series data with dependencies across time steps. Proper preprocessing, sequence generation, and hyperparameter tuning significantly enhance performance. LSTM proves to be highly effective for real-world forecasting applications such as environmental monitoring.